Practice workbook · Hormones and sleep: the pineal gland and melatonin

The pineal gland's melatonin release in response to light as an internal influence on sleep.

5 question parts: 1 worked-example retry and 4 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: PSY-SLP-03

1. Question 30 · part whole · 2 marks

2024 JUN · Paper 02

Worked-example retry · Attempt part whole only.

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30 Describe how the pineal gland and melatonin influence a person's sleep.

(Total for Question 30 = 2 marks)

Working and answer

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2. Question 31 · part whole · 2 marks

2020 NOV · Paper 02

Further practice · Attempt part whole only.

31 Pragya struggles to wake in the mornings and can feel sleepy throughout the day. She has stopped socialising with her friends because she is too tired to go out. Pragya visits the doctor who thinks there may be a problem with her pineal gland.

Explain how the pineal gland may be affecting Pragya's sleep.

(Total for Question 31 = 2 marks)

DO NOT WRITE IN THIS AREA

Working and answer

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3. Question 34 · part whole · 4 marks

2021 NOV · Paper 02

Further practice · Attempt part whole only.

34 Paolo is having difficulty getting to sleep and is often awake at night. His doctor recommends that two hours before going to sleep Paolo should dim the lights and stop using his mobile phone because it has a bright screen.

When Paolo followed the recommendations, he slept better. He then started using the mobile phone at night, including when he went to bed. Paolo's sleep problems have returned.

Explain **two** ways that the role of hormones could account for Paolo's sleep.

You must use concepts or research evidence to justify your answer.

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(Total for Question 34 = 4 marks)

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3. Question 34 · part whole · 4 marks · Working and answer

Working and answer

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4. Question 33 · part whole · 9 marks

2022 JUN · Paper 02

Further practice · Attempt part whole only.

***33** Basil is having problems with sleeping. He is finding it difficult to go to sleep at night and struggling to wake in the morning. Basil visited his doctor who tested the functioning of his pineal gland. The doctor also thought that Basil may have problems with the hormone levels that affect sleep.

Basil was given medication that helps regulate sleep hormones. When Basil takes the medication, he finds he can get to sleep better but he still finds it hard to wake up in the morning.

Assess how well hormones can explain the problems Basil has with sleeping.

(9)

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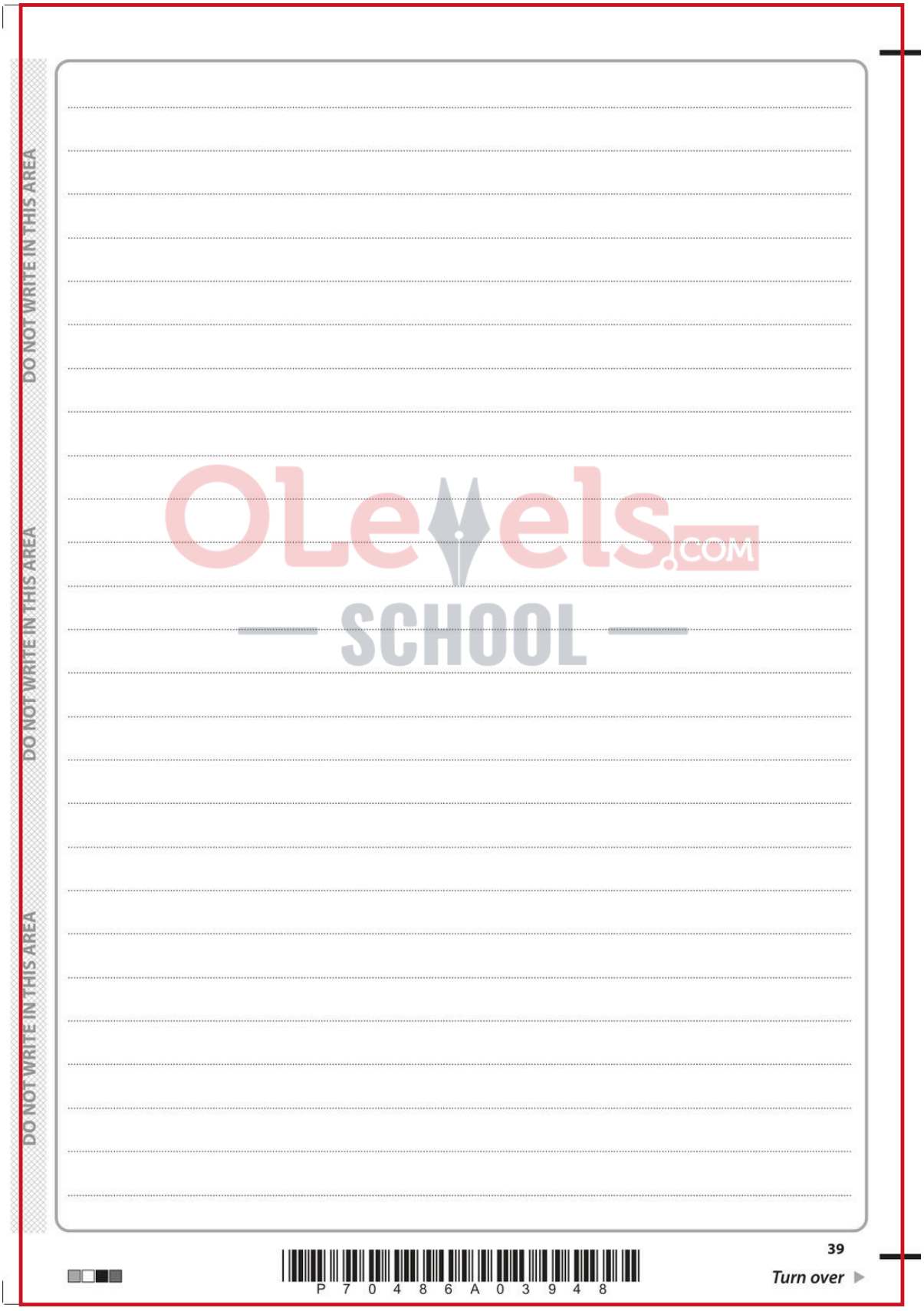
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4. Question 33 · part whole · 9 marks

2022 JUN · Paper 02

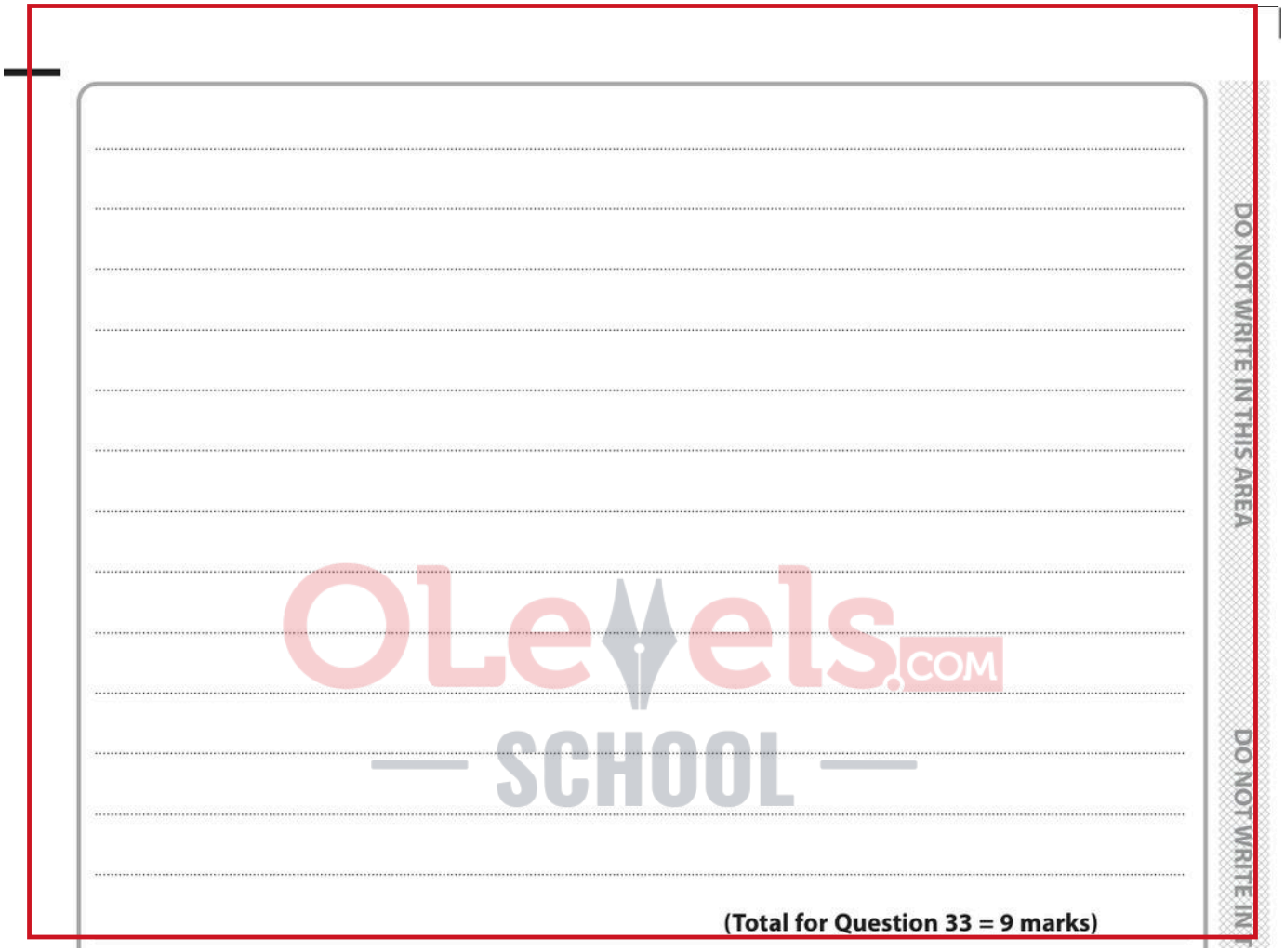
Further practice · Attempt part whole only.



4. Question 33 · part whole · 9 marks

2022 JUN · Paper 02

Further practice · Attempt part whole only.



4. Question 33 · part whole · 9 marks · Working and answer

Working and answer

OLevels.com SCHOOL

5. Question 32 · part whole · 4 marks

2023 JUN · Paper 02

Further practice · Attempt part whole only.

32 Tilly is struggling with her sleep. She finds it hard to go to sleep at night and sometimes sleeps until 11am. Tilly feels tired a lot of the time and is worried that her grades at school will suffer because of her lack of sleep.

Tilly visits her doctor to get help with her sleep. The doctor decides to test her hormone levels.

Explain **one** strength and **one** weakness of using hormones to explain Tilly's sleep problems.

Strength

Weakness

(Total for Question 32 = 4 marks)

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5. Question 32 · part whole · 4 marks · Working and answer

Working and answer

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